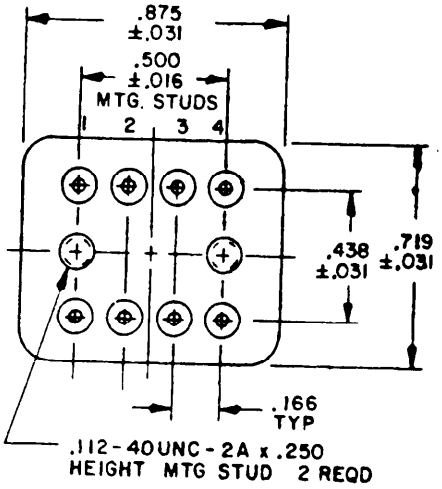
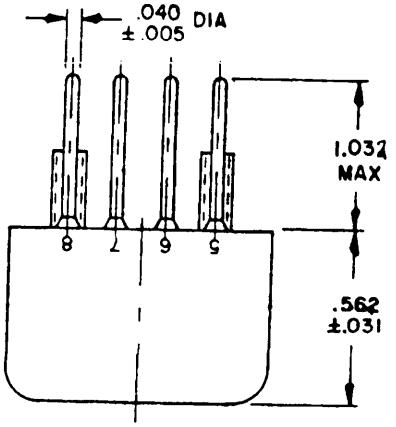
MILITARY SPECIFICATION SHEET

TRANSFORMER, AUDIO FREQUENCY, TF4R21YY

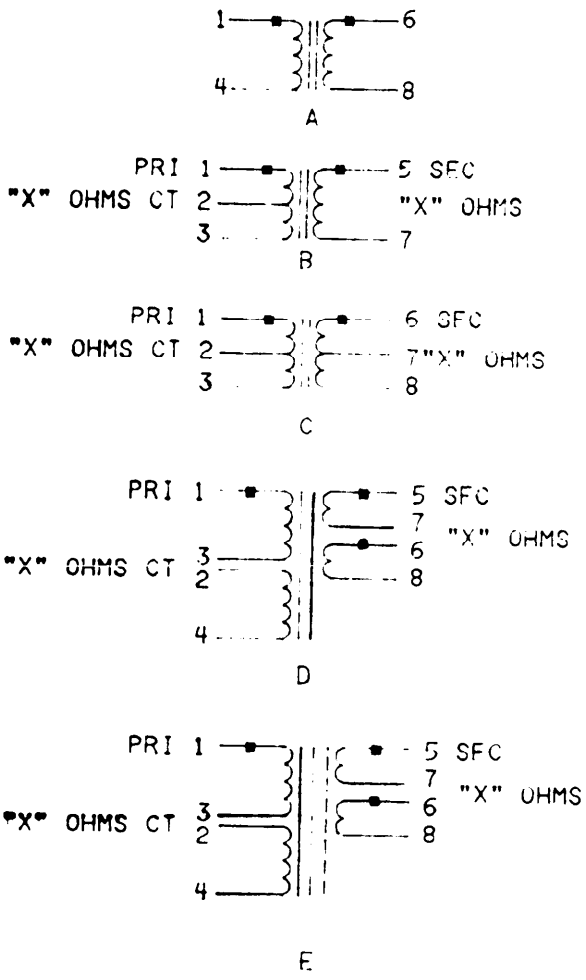
This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the transformers described herein shall consist of this specification and the latest issue of Specification MIL-T-27.





INCHES	MM
.005	.13
.016	.41
.031	.79
.040	1.02
.112	2.84
.166	4.22
.250	6.35
.438	11.13
.500	12.70
.562	14.27
.719	18.26
.875	22.22
1.032	26.19



WORKING VOLTAGE : 175 VOLTS

CIRCUIT DIAGRAMS AND MARKING

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. Marking shall be on the side of the case.
4. Primary and secondary electrical values shall be marked as specified in table I, as applicable.

FIGURE 1. Dimensions and configuration.

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FSC 5950

REQUIREMENTS: (When numbers in parentheses, i.e. (1-2) are used, they indicate the windings and the extreme terminals of the windings.)

Electrical ratings: See table I.

Working voltage: 175 volts.

Frequency range: Shall be as follows:

Dash no.	frequency range (Hz)
-02	150 to 10,000
-01, 03 thru 10	300 to 10,000
-11 thru 14	300 to 20,000

Design and construction:

Dimensions and configuration: See figure 1.

Duty cycle: Continuous.

Case: Metal.

Terminals: Pin type.

Diameter: .040 inch ±0.005.

Length: 1.032 inches maximum.

Weight: 1 ounce, maximum.

Operating temperature range: -55° to +105°C.

Terminal strength: Method 211 of MIL-STD-202, test condition A, 2.0 pounds.

Dielectric withstanding voltage (each winding):

At sea level: 500 volts rms.

Electrical characteristics:

No load (center tap unbalance): ±1 percent at 1 volt, 5 kHz across the primary.

Harmonic distortion: Total harmonic content of the output shall not exceed 6.5 percent maximum with specified output power and primary dc current at 300 Hz (see table I).

Insertion loss: At the specified power level (see table I), the insertion loss shall be a maximum of 2.0 dB at 1 kHz.

Frequency response: ±2 dB at the rated source and load impedances (see table II) with specified output power (see table I) over the specified frequency range (see table II) using a reference frequency of 1 kHz.

Polarity: Additive, with terminals connected as specified in table II.

Electrostatic shield: For those parts with electrostatic shielding (circuit diagram E), voltage ratio shall be 2 to 1 at 20 kHz.

Vibration (high frequency): MIL-STD-202, method 204, test condition D.

Marking location: See figure 1.

Part number: M27/200 - (dash number from table I).

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TABLE I. Electrical ratings.

Dash no.	Circuit diagram	Primary impedance (ohms)	Secondary impedance (ohms)	Primary dc current (mA)	Power level (mW)	Primary dc resistance ±25°C (ohms)	Secondary dc resistance ±25°C (ohms)
01	A	30K (1-4)	50 (6-8)	1	30	2875	4.6
02	A	50 (1-4)	62.5K (6-8)	0	20	13.4	3600
03	A	10K (1-4)	90K (6-8)	0	20	710	3170
04	A	25K (1-4)	1K (6-8)	.5	10	800	110
05	B	500 CT (1-3)	50 (6-8)	3.5	30	50	5
06	C	10K CT (1-3)	1.2K CT (6-8)	2	30	540	85
07	C	10K CT (1-3)	2K CT (6-8)	2	30	600	166
08	C	500 CT (1-3)	600 CT (6-8)	2	30	26	70
09	C	1.5K CT (1-3)	600 CT (6-8)	10	30	69	60
10	C	10K CT (1-3)	10K CT (6-8)	7	30	560	650
11	D	500/125 SPLIT (1-4)	5000/1250 SPLIT (5-8)	1	100	20	290
12	D	5000/1250 SPLIT (1-4)	5000/1250 SPLIT (5-8)	12	50	200	275
13	E	10K CT/2.5K SPLIT (1-4)	200 CT/50 SPLIT (5-8)	4	50	490	16
14	E	20K CT/5K SPLIT (1-4)	800 CT/200 SPLIT (5-8)	2	100	733	82
				1	100		

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TABLE II. Electrical characteristics.

Dash no.	Frequency response: ±2 dB			Polarity: Additive with terminals (below) connected.
	Z_S (ohms)	Z_L (ohms)	Frequency range (Hz)	
01	30K (1-4)	50 (6-8)	300 to 10,000	4 and 6
02	50 (1-4)	62.5K (6-8)	150 to 10,000	↓
03	10K (1-4)	90K (6-8)	300 to 10,000	↓
04	25K (1-4)	1K (6-8)	↓	↓
05	500 (1-3)	50 (5-7)	↓	3 and 5
06	10K (1-3)	1.2K (6-8)	↓	3 and 6
07	10K (1-3)	2K (6-8)	↓	↓
08	500 (1-3)	600 (6-8)	↓	↓
09	1.5K (1-3)	600 (6-8)	↓	↓
10	10K (1-3)	10K (6-8)	↓	↓
11	500 (1-4)	5000 (5-8)	300 to 20,000	2 and 3; 4 and 5; 6 and 7
12	5000 (1-4)	5000 (5-8)	↓	↓
13	10K (1-4)	200 (5-8)	↓	↓
14	20K (1-4)	800 (5-8)	↓	↓

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - MI
Navy - OS, SH
Air Force - 17, 99
DLA - ES

User activities:
Army - AR, ME
Navy - AS, MC
Air Force - 19

Agent:
DLA - ES

Preparing activity:
Army - ER

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